

Aerodynamics, Turbulence, and Computational Fluid Dynamics

How air produces lift and drag, why flows become turbulent, and how CFD turns governing equations into predictions

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① **Governing physics**

Mass and momentum conservation, viscosity, and the Reynolds number.

② **Aerodynamic forces**

Where lift and drag actually come from.

③ **Instability and turbulence**

Boundary layers, separation, shedding, and the energy cascade.

④ **Computational fluid dynamics**

Meshing, discretization, turbulence models, and how to trust a simulation.

The fluid as a continuum

Air contains an enormous number of molecules, so tracking individual molecular trajectories is neither useful nor necessary for engineering-scale flows.

Instead, we describe the flow using smooth fields:

$$\rho(\boldsymbol{x}, t), \quad p(\boldsymbol{x}, t), \quad \boldsymbol{u}(\boldsymbol{x}, t), \quad T(\boldsymbol{x}, t).$$

The continuum approximation is appropriate when the length scales of interest are much larger than the molecular mean free path.

Two viewpoints

- Eulerian: observe the flow at fixed locations in space.
- Lagrangian: follow individual fluid parcels.

CFD is fundamentally an Eulerian description: the governing equations are solved at locations associated with a computational mesh.

Conservation of mass

Mass is neither created nor destroyed. For a differential control volume,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

- $\partial \rho / \partial t$: local accumulation of mass.
- $\nabla \cdot (\rho \mathbf{u})$: net mass flux leaving the volume.

For steady, incompressible flow through a duct,

$$A_1 V_1 = A_2 V_2.$$

Thus a smaller cross-sectional area requires a larger velocity when density is constant.

Newton's second law for a fluid parcel

For an incompressible Newtonian fluid with constant viscosity,

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}$$

Acceleration

$$\frac{D\mathbf{u}}{Dt} = \frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u}$$

The first term is local acceleration; the second is convective acceleration.

A steady flow can still accelerate a fluid parcel: a parcel moving into a region of increasing velocity has nonzero convective acceleration.

Forces

- $-\nabla p$: pressure force
- $\mu \nabla^2 \mathbf{u}$: viscous diffusion of momentum
- $\rho \mathbf{g}$: body force

The Reynolds number: inertia versus viscosity

$$\text{Re} = \frac{\rho U L}{\mu} = \frac{U L}{\nu}$$

The Reynolds number compares the characteristic magnitude of inertial effects with viscous effects.

Low Re

Viscosity dominates. Motion is strongly damped.

Intermediate Re

Instabilities can grow.

High Re

Inertia dominates and turbulent motion becomes possible.

The same Navier–Stokes equations govern all three regimes; the relative importance of the terms changes with the nondimensional parameters.

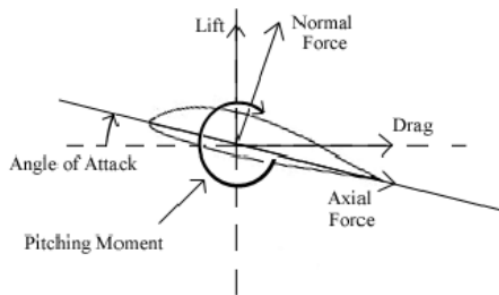
Where lift and drag come from

Aerodynamic forces arise from two surface stresses:

- **Pressure:** acts normal to the surface.
- **Viscous shear:** acts tangentially to the surface.

Lift is usually dominated by pressure differences.

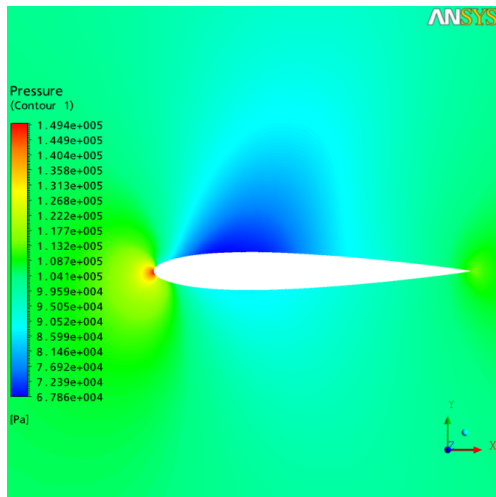
Drag contains both pressure/form drag and skin-friction drag.



$$L = \frac{1}{2} \rho V_{\infty}^2 S C_L$$

$$D = \frac{1}{2} \rho V_{\infty}^2 S C_D$$

The pressure coefficient along the chord



A convenient nondimensional measure of pressure is

$$C_p = \frac{p - p_\infty}{\frac{1}{2}\rho V_\infty^2}$$

What the CFD field is showing

- A lifting airfoil develops a strong suction peak on the upper surface near the leading edge.
- The upper and lower surfaces sit at different pressure levels.
- Integrating the pressure and shear stresses over the airfoil gives the aerodynamic force.

Source: ANSYS CFD pressure field, from the supplied corrected presentation.

Lift as a push on the air



Lift can also be understood from momentum conservation. Because the wing turns the flow, a control volume around it sees different incoming and outgoing momentum fluxes:

$$\mathbf{F}_{\text{momentum}} = \oint_{CS} \rho \mathbf{u} (\mathbf{u} \cdot \mathbf{n}) dA$$

The air leaves with a net **downward momentum**, so by Newton's third law the wing experiences an equal and opposite upward force.

Coloured smoke makes that wake visible here: the flow behind the wing is driven downward and rolls up into a trailing vortex.

Circulation and the Kutta–Joukowski theorem

Circulation is the line integral of velocity around a closed contour:

$$\Gamma = \oint_C \mathbf{u} \cdot d\boldsymbol{\ell}$$

For two-dimensional inviscid flow,

$$L' = \rho V_{\infty} \Gamma$$

where L' is lift per unit span.

The Kutta condition requires the flow to leave a sharp trailing edge smoothly. This determines the circulation associated with the lifting solution.

Key idea: circulation is an integrated property of a velocity field; it is not the same thing as local vorticity.

Vorticity: local spin in the flow

Vorticity is defined by

$$\boldsymbol{\omega} = \nabla \times \boldsymbol{u}$$

It is a **local** property of the velocity field. By contrast,

$$\Gamma = \oint_C \boldsymbol{u} \cdot d\boldsymbol{\ell}$$

is an **integrated** property over a closed contour.

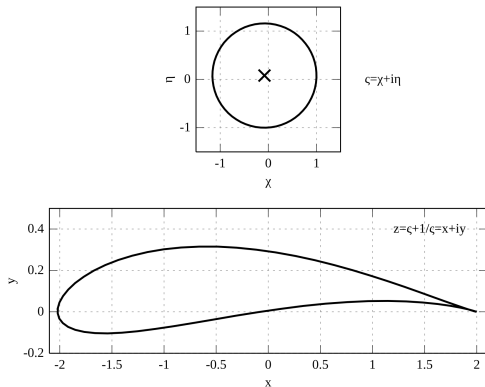
- Vorticity can be concentrated in a thin boundary layer.
- Vorticity can be concentrated in a wingtip vortex core.
- A region can have strong local rotation without the same circulation being present on every surrounding contour.

This distinction becomes essential when interpreting CFD visualizations of vortical structures.

Ideal flow and the Joukowski transformation

The Joukowski transformation maps a circle in the complex plane into an airfoil:

$$z = \zeta + \frac{a^2}{\zeta}$$



The usefulness of the transformation is conceptual: a problem with a simple geometry and a known potential-flow solution such as flow past a circular cylinder can be mapped into the corresponding airfoil flow.

Moving the center of the circle in the ζ -plane changes the thickness and camber of the resulting section.

Bernoulli's equation

For steady, inviscid, incompressible flow along a streamline,

$$p + \frac{1}{2}\rho V^2 + \rho g z = \text{constant}$$

Thus, along a streamline,

$$V \uparrow \implies p \downarrow$$

Important qualification: Bernoulli's equation is not an independent law of nature. It follows from the momentum equation after imposing the appropriate assumptions.

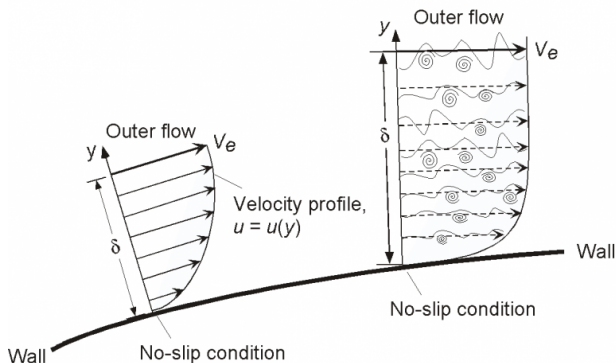
The common “equal transit time” explanation for lift is not correct. The actual velocity field around a lifting wing does not require equal travel times over the upper and lower surfaces.

Boundary layers

The no-slip condition requires $u = 0$ at a stationary wall, so viscous effects become important in a thin region adjacent to the surface.

(a) Laminar boundary layer

(b) Turbulent boundary layer



Laminar boundary layer

- Smooth, ordered velocity profile.
- Relatively little mixing across the layer.

Turbulent boundary layer

- Strong fluctuations and mixing.
- Fuller velocity profile.

Boundary-layer thickness and the near-wall region

For a classical zero-pressure-gradient laminar boundary layer, the characteristic scaling is

$$\boxed{\delta(x) \sim \frac{x}{\sqrt{\text{Re}_x}}} \quad \text{Re}_x = \frac{U_\infty x}{\nu}.$$

Equivalently,

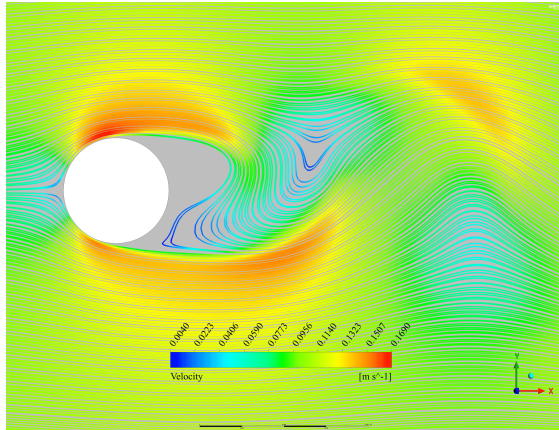
$$\frac{\delta}{x} \sim \text{Re}_x^{-1/2}.$$

The boundary layer carries a momentum deficit relative to the external flow.

That momentum deficit becomes crucial when the pressure gradient is adverse: the near-wall fluid is the first part of the flow that can lose enough momentum to reverse direction.

Adverse pressure gradients and separation

An adverse pressure gradient, $\boxed{dp/dx > 0}$ for flow moving in the positive x direction, means the pressure rises downstream and opposes the motion.



The wall shear stress is

$$\boxed{\tau_w = \mu \left. \frac{\partial u}{\partial y} \right|_{wall}}$$

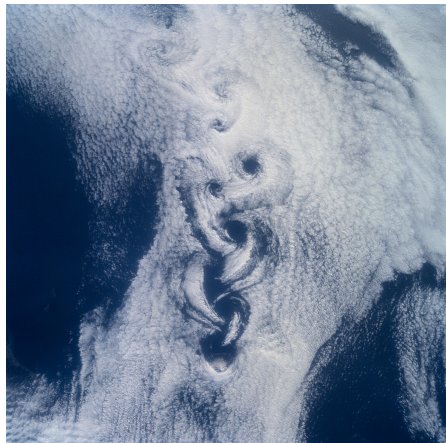
and it vanishes at separation, $\boxed{\tau_w \rightarrow 0}$, changing sign downstream of the separation point.

The boundary layer then detaches from the surface, leaving a recirculating region and often a large wake.

Source: ANSYS CFD velocity and streamline visualization, from the supplied corrected presentation.

Vortex shedding

Behind a bluff body the separated shear layers become unstable: a small asymmetry grows, one side sheds a vortex, and the pressure field then promotes shedding from the other side.



The process repeats at a well-defined frequency:

$$\text{St} = \frac{fD}{U}$$

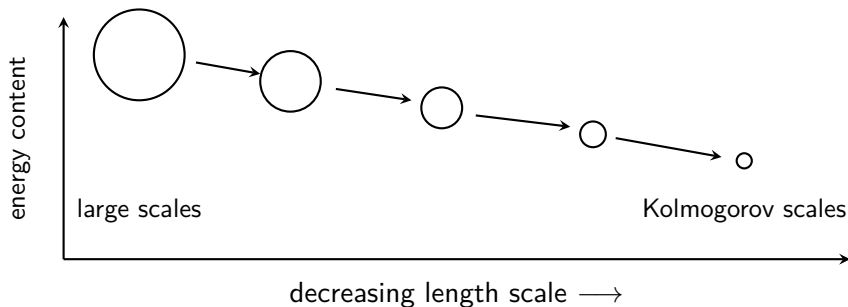
where f is the shedding frequency, D a characteristic body dimension, and U the incoming velocity.

The wake is a staggered double row of counter-rotating vortices: a von Kármán vortex sheet. Here an island is the bluff body and the cloud deck is the tracer.

Source: Von Kármán vortices off Rishiri Island, Japan, NASA (public domain).

From instability to turbulence: the energy cascade

Turbulence contains a broad range of spatial and temporal scales.



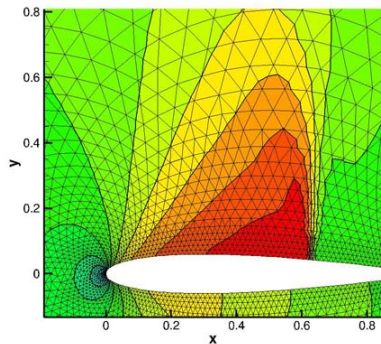
Energy is introduced at large scales, transferred through progressively smaller scales at a nearly constant rate, and ultimately dissipated by viscosity at the smallest scales.

This multi-scale structure is one reason turbulence is so expensive to resolve directly.

Why we need CFD, and what a mesh does

Realistic aerodynamic flows are nonlinear, geometrically complicated, and contain many interacting scales, so CFD replaces the continuous field problem with a discrete one:

continuous equations $\longrightarrow$ mesh + discrete equations $\longrightarrow$ numerical solution.



A mesh divides the domain into cells. Anything smaller than the numerical resolution cannot be represented accurately.

Resolution should therefore be concentrated where the physics demands it:

- near walls,
- in shear layers and wakes,
- around shocks or strong gradients.

Source: Airfoil CFD mesh and solution field, from the supplied corrected presentation.

The finite-volume method

The finite-volume method starts from an integral conservation law:

$$\frac{d}{dt} \int_V U dV + \oint_{\partial V} \mathbf{F} \cdot \mathbf{n} dA = \int_V S dV$$

For each control volume:

- **Accumulation:** change of conserved quantity stored inside the cell.
- **Flux:** transport through the cell faces.
- **Source:** local production or removal.

The key advantage is conservation: what leaves one cell through a shared face enters its neighbour through the same face with the opposite sign.

The result is a large algebraic system that the numerical solver must converge.

A CFD solution is more than a colorful contour

A visually smooth contour does not guarantee a physically correct solution.

A trustworthy CFD result requires several layers of scrutiny:

- ① **Spatial discretization:** is the mesh sufficiently resolved?
- ② **Temporal discretization:** is the time step appropriate?
- ③ **Iterative convergence:** have the nonlinear/algebraic equations converged?
- ④ **Boundary conditions:** do they represent the physical problem?
- ⑤ **Physical modelling:** are the assumptions appropriate?
- ⑥ **Turbulence modelling:** is the closure appropriate?

converged $\nRightarrow$ correct

The closure problem

Averaging the Navier–Stokes equations introduces correlations between velocity fluctuations. The Reynolds stress tensor appears as

$$-\rho \overline{u'_i u'_j}$$

These are additional unknowns, producing the turbulence closure problem.

DNS

All relevant scales are resolved directly. Very expensive.

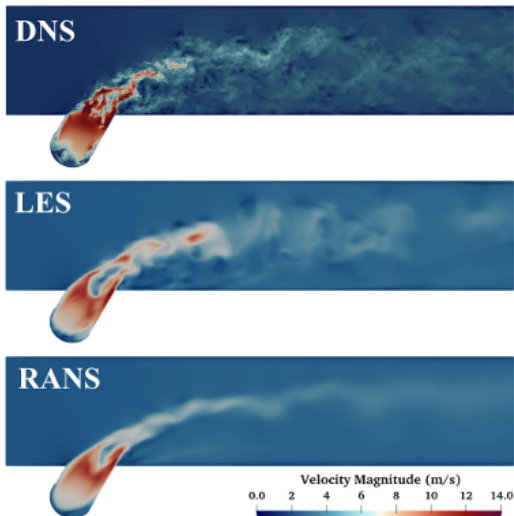
LES

Large scales are resolved; smaller scales are modelled.

RANS

The turbulent fluctuations are not resolved; their net effect is modelled.

Visual Comparison of DNS, LES and RANS



Eddy-viscosity models

A common closure uses the Boussinesq hypothesis:

$$-\rho \overline{u'_i u'_j} \approx 2\mu_t S_{ij} - \frac{2}{3}\rho k \delta_{ij}$$

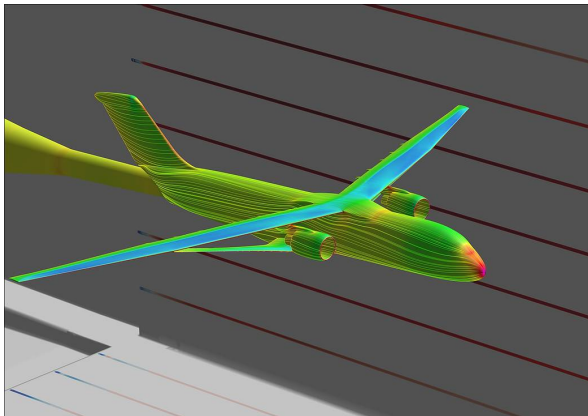
where μ_t is the turbulent or eddy viscosity.

Unlike molecular viscosity μ , μ_t is not a material property. It varies with the local flow and must be obtained from the turbulence model.

- $k-\epsilon$: historically robust and effective for many free-shear flows.
- $k-\omega$: naturally formulated for the near-wall region.
- **SST** $k-\omega$: blends the advantages of the two formulations and is widely used for aerodynamic flows with possible separation.

CFD visualization: seeing the computed flow

CFD lets us visualize quantities that are difficult or impossible to measure everywhere experimentally.



Examples include velocity magnitude, pressure, Mach number, vorticity, turbulent kinetic energy, surface pressure coefficient, and streamlines or pathlines.

Here the aircraft surface is coloured by pressure and the traces on it are surface streamlines.

But a visualization is only useful when we understand what quantity is being plotted and how it was computed.

Source: LAVA simulation of the transonic truss-braced wing, NASA/Ames, Daniel Maldonado (public domain).

Verification and validation

Verification asks:

Did we solve the equations correctly?

Typical checks:

- mesh refinement,
- time-step refinement,
- residual convergence,
- conservation checks,
- comparison with analytical or manufactured solutions.

Validation asks:

Do the equations and models represent reality?

Validation requires comparison with trusted experimental measurements or other appropriate reference data.

A simulation can be perfectly verified and still fail validation.

CFD as a design tool

Forward analysis asks:

$$\boxed{\text{geometry} \longrightarrow \text{flow}}$$

Design reverses the direction:

$$\boxed{\text{desired flow} \longrightarrow \text{geometry}}$$

This is an inverse problem. A generic design objective can be written as

$$J = J(\text{geometry}, \text{flow solution}),$$

and optimization seeks a geometry that minimizes or maximizes J subject to physical and manufacturing constraints.

For a functional

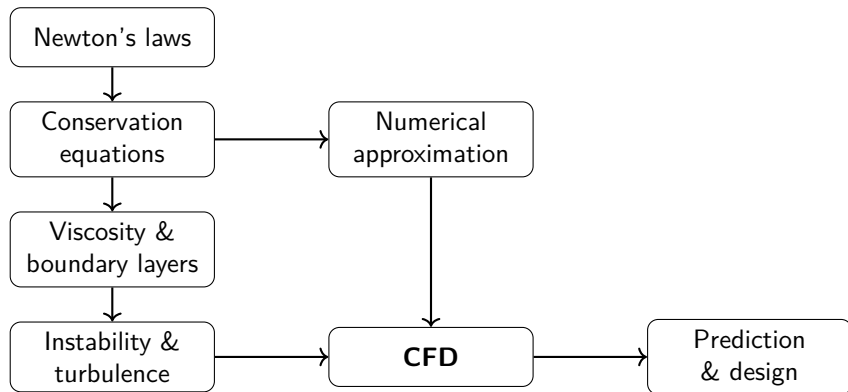
$$J[y] = \int_a^b F(x, y, y') dx,$$

the Euler–Lagrange equation is

$$\boxed{\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.}$$

In practice, gradient-based shape optimization uses adjoint methods, which make the cost of one gradient nearly independent of the number of design variables.

The central picture



The fundamental question is not “Can we simulate it?”

It is “Do we understand what the simulation is actually telling us?”

Turbulence modelling

- Hybrid RANS–LES: $\ell_{DES} = \min(\ell_{RANS}, C_{DES}\Delta)$
- ML-augmented closures: learned correction $\beta(\mathbf{x})$ on production terms
- Physics-informed neural networks (PDE residual in the loss)

High-fidelity methods

- Wall-modelled LES (WMLES) toward industrial Re
- High-order DG / spectral element methods
- Immersed boundary / overset meshing for complex, moving geometry

Design & optimization

- Adjoint-based shape optimization:

$$\left(\frac{\partial R}{\partial u}\right)^T \psi = -\frac{\partial J}{\partial u}, \quad \frac{dJ}{d\alpha} = \frac{\partial J}{\partial \alpha} + \psi^T \frac{\partial R}{\partial \alpha}$$

- Topology optimization for fluid domains

Uncertainty & reduced-order modeling

- UQ: turbulence model-form & BC uncertainty propagation
- ROM: $u(\mathbf{x}, t) \approx \sum_{i=1}^r a_i(t) \phi_i(\mathbf{x})$ (POD/DMD)

Multiphysics & compute scale

- Fluid–structure interaction (aeroelasticity)
- Conjugate heat transfer / multiphase flow
- GPU / exascale CFD toward DNS at industrial Re